

**MARK SCHEME for the October/November 2009 question paper
for the guidance of teachers**

9702 PHYSICS

9702/21

Paper 21 (AS Structured Questions), maximum raw mark 60

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

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Page 2	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE A/AS LEVEL – October/November 2009	9702	21

- 1 (a) (i) car uses $210 / 14 = 15$ litres of fuel C1
volume reading = 45 litres A1 [2]
- (ii) from 'full' to '3/4' mark B1 [1]
- (b) (i) line/graph does not pass through ('empty, 0') / there is an intercept B1 [1]
(do not allow 'non-linear')
- (ii) (meter shows zero fuel when there is some left in the tank so)
acts as a 'reserve' B1 [1]

[Total: 5]

- 2 (a) (i) (air) resistance increases with speed M1
resultant / accelerating force decreases A1 [2]
- (ii) either (air) resistance is zero
or weight / gravitational force is only force B1 [1]
- (b) use of gradient of a tangent M1
acceleration = $1.9 \pm 0.2 \text{ m s}^{-2}$ A2 [3]
(for values $> \pm 0.2$ but ≤ 0.4 , allow 1 mark)
(answer 3.3 m s^{-2} scores no marks)
- (c) (i) 1 weight = $90 \times 9.8 = 880 \text{ N}$ A1 [1]
(use of $g = 10 \text{ m s}^{-2}$ then deduct mark but once only in the Paper)
2 accelerating force = $90 \times 1.9 = 170 \text{ N}$...(allow ecf) A1 [1]
- (ii) resistive force = $880 - 170 = 710 \text{ N}$ A1 [1]
(allow ecf but only if resistive force remains positive)

[Total: 9]

- 3 (a) (i) either sum / total momentum (of system of bodies) is constant
or total momentum before = total momentum after M1
for an isolated system / no (external) force acts on system A1 [2]
- (ii) zero momentum before / after decay M1
so α -particle and nucleus D must have momenta in opposite directions A1 [2]
- (b) (i) kinetic energy = $\frac{1}{2} mv^2$ C1
 $1.0 \times 10^{-12} = \frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times v^2$ M1
 $v = 1.7 \times 10^7 \text{ m s}^{-1}$ A0 [2]
- (ii) $1.7 \times 10^7 \times 4u = 216u \times V$ C1
 $V = 3.1 \times 10^5 \text{ m s}^{-1}$ A1 [2]
(accept $3.2 \times 10^5 \text{ m s}^{-1}$, do not accept 220 rather than 216)

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE A/AS LEVEL – October/November 2009	9702	21

- (c) $(1.7 \times 10^7)^2 = 2 \times \text{deceleration} \times 4.5 \times 10^{-2}$ C1
 $\text{deceleration} / a = 3.2 \times 10^{15} \text{ m s}^{-2}$ A1 [2]
 (accept calculation based on calculating $F = 2.22 \times 10^{-11} \text{ N}$
 and then use of $F = ma$)

[Total: 10]

- 4 (a) (i) returns to original shape / size / length etc. B1
 when load / distorting forces / weight / strain is removed B1 [2]
 (ii) 1 $R = \rho L / A$ B1 [1]
 2 $E = WL / Ae$ B1 [1]

- (b) $E = WR / e\rho$ C1
 $= (34 \times 0.44) / (7.7 \times 10^{-4} \times 9.2 \times 10^{-8})$ C1
 $= 2.1 \times 10^{11} \text{ Pa}$ A1 [3]

[Total: 7]

- 5 (a) transfer / propagation of energy M1
 as a result of oscillations / vibrations A1 [2]
 (b) (i) displacement / velocity / acceleration (of particles in the wave) B1 [1]
 (ii) displacement etc. is normal to direction of energy transfer /
 travel of wave / propagation of wave(not 'wave motion') B1 [1]
 (iii) displacement etc. along / same direction of energy transfer /
 travel of wave / propagation of wave(not 'wave motion') B1 [1]
 (c) diffraction: suitable object, means of observation M1
 either laser or lamp and aperture M1
 or distant source A1
 light region where darkness expected A1
 interference: suitable object, means of observation and illumination B1
 light and dark fringes observed B1
 appropriate reference to a dimension for diffraction or
 for interference B1 [6]

[Total: 11]

- 6 (a) energy transferred from source / changed from some form to electrical M1
 per unit charge (to drive charge round a complete circuit) A1 [2]
 (b) and power in $R = I^2 X$ M1
 $E = I(X + r)$ M1
 power in cell = EI and algebra clear leading to ratio = $X / (X + r)$ A1 [3]

Page 4	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE A/AS LEVEL – October/November 2009	9702	21

(c) (i) 1.4 W A1
0.40 Ω (allow $\pm 0.05 \Omega$) A1 [2]

(ii) current in circuit = $\sqrt{1.4/0.4} = 1.87 \text{ A}$ C1
1.5 = 1.87 ($r + 0.40$) C1
 $r = 0.40 \Omega$ A1 [3]

(d) either less power lost / energy wasted / lost
or greater efficiency (of energy transfer) B1 [1]

[Total: 11]

7 (a) deviation shown correctly B1 [1]

(b) smaller deviation (not zero deviation) M1
acceptable path wrt position of N A1 [2]

(c) the nucleus is (very) small M1
in comparison to the atom A1 [2]
(special case: 'atom is mostly empty space' scores 1 mark)

(d) deviation depends on charge on the nucleus / N / electrostatic repulsion B1
same charge so no change in deviation B1 [2]

[Total: 7]